Time-Domain Imaging of Rotating, Cloud-Covered Exoplanets with the Solar Gravitational Lens: Reconstruction Algorithms, Observing-Cadence Requirements, and Mission Targets

Sanskar Sontakke
Independent Researcher
sanskarsontakke@gmail.com ORCID: 0009-0008-4531-5707

ABSTRACT

The solar gravitational lens (SGL) can, in principle, deliver multipixel images of terrestrial exoplanets from heliocentric distances $z \simeq 650$–900 AU. In practice the planet rotates and its cloud cover evolves while a spacecraft raster-scans the kilometer-scale image cylinder one "pixel" at a time, so the measurements are temporally aliased; recent studies identify this dynamic inversion problem as a leading unsolved challenge for the concept. We present an open, end-to-end simulation framework that reproduces the published SGL photometric benchmarks (the aperture-averaged $d/4\rho$ kernel; per-sample $\mathrm{SNR_C} = 43.16$ for 1800 s dwells on an exo-Earth at 30 pc from 650 AU; the $0.891\,D/(d\sqrt{N})$ deconvolution penalty) and then treats rotation, orbital illumination, and stochastic, advecting clouds explicitly. We formulate image recovery as a regularized generalized least-squares inversion of the time-tagged sample stream (time-domain inversion, TDI), in which rotation is handled exactly by the forward operator and cloud variability enters through its temporal covariance. For a rotating, cloud-free planet the method recovers the surface albedo map with Pearson $r = 0.99$ (SSIM $= 0.92$), removing the SNR $\lesssim 2$ limitation reported for prior dynamic reconstructions. Clouds, not coronal photon noise, dominate the error budget: the per-sample cloud-induced fluctuation exceeds coronal shot noise by factors of 14–23, and at fixed observing resources image fidelity collapses from $r = 0.99$ to $r \simeq 0.26$ as the mean cloud cover rises from 0 to 72%. We derive an observing-cadence law: at fixed photon budget and fixed 90-day wall-clock time, replacing 4 long dwells per pixel with 64 short revisits raises r from 0.15 to 0.58, because revisits sample independent cloud states. Reconstruction requires the rotation period a priori to $\sim 10^{-4}$. Finally, we map these requirements onto a mission: we rank the five nearest confirmed potentially habitable planets as SGL targets, give the celestial placement of each focal line, quantify per-target image-plane dynamics (tracking budgets of 0.1–5.8 $\mathrm{km\,s^{-1}}$ per 90 days), and summarize the propulsion (deep-perihelion sailcraft with $\sigma_{\mathrm{tot}} \simeq 2.3$–4.9 $\mathrm{g\,m^{-2}}$), power (radioisotope tiles), and downlink resources needed at $z \gtrsim 550$ AU. All simulations, figures, and tables are reproducible from the included code.

Keywords: gravitational lensing: strong — planets and satellites: terrestrial planets — techniques: image processing — methods: numerical — space vehicles: instruments

1 Introduction

Resolving the surface of an Earth-like exoplanet is beyond any credible filled-aperture or interferometric architecture: a 10×10 surface map of an exo-Earth at 10 pc requires resolution elements of $\sim$0.85 μas, i.e., a $\sim 10^2$ km optical aperture, and integration times for conventional concepts reach 10^4–10^5 yr once realistic backgrounds are included (Turyshev 2026a). The solar gravitational lens (SGL) evades this scaling by using the Sun as the primary optical element (Eshleman 1979; Turyshev & Toth 2017). Wave-optical treatments show that a strong-interference region forms beyond $z_0 \simeq b^2/2r_g \simeq 547.8$ AU (for rays of impact parameter $b \simeq R_\odot$, with $r_g = 2GM_\odot/c^2 \simeq 2.95$ km), with on-axis amplification $\mu_0 = 4\pi^2 r_g/\lambda \simeq 1.2 \times 10^{11}$ at $\lambda = 1$ μm and angular resolution of order 10^{-10} arcsec (Turyshev & Toth 2017, 2019). An Earth-radius planet at 30 pc is compressed into an image cylinder only $D_{\mathrm{img}} \simeq 1.34$ km across at $z = 650$ AU; a meter-class telescope acts as a single-pixel detector that must physically traverse the image plane, measuring the integrated brightness of the planetary Einstein ring at each sampling location and reconstructing the source by deconvolution (Turyshev & Toth 2020; Toth & Turyshev 2021; Turyshev & Toth 2022a).

The optical side of this problem is now well characterized. The aperture-averaged point-spread function falls off as $d/(4\rho)$, so each sample mixes light from the entire planet; removing this blur incurs a deconvolution penalty $\mathrm{SNR_R}/\mathrm{SNR_C} \simeq 0.891\,D/(d\sqrt{N})$ that is strongly mitigated by wide pixel spacing (Toth & Turyshev 2021; Turyshev & Toth 2022a). The solar corona is the dominant photon background; for an exo-Earth at $z_0 = 30$ pc observed from $z = 650$ AU with a $d = 1$ m telescope at $\lambda = 1$ μm, the signal and corona rates are $Q_{\mathrm{exo}} = 8.01 \times 10^4$ $\mathrm{s^{-1}}$ and $Q_{\mathrm{cor}} = 6.20 \times 10^9$ $\mathrm{s^{-1}}$, giving a convolved-image signal-to-noise ratio $\mathrm{SNR_C} = 43.2$ per 1800 s dwell (Turyshev & Toth 2022a; Turyshev 2026b). Megapixel-class static imaging within mission-compatible integration times follows (Toth & Turyshev 2021; Turyshev & Toth 2022a), and independent analyses of ring-based approaches (Madurowicz & Macintosh 2022) sharpened the role of coronal statistics in these budgets.

The unsolved problem is temporal. The planet rotates and orbits, its illumination changes, and—most seriously—its cloud cover reorganizes on day timescales while the raster is acquired over weeks to months. Toth & Turyshev (2023) performed the first dynamic reconstructions of a rotating, orbiting planet and recovered topography only at SNR $\lesssim 2$. Toth (2025) added a stochastic cloud model, treated the clouds as noise to be averaged out (a "long-exposure" philosophy), and found that even 7–14% cloud cover degrades the recovered SNR below unity, concluding with "a cautious assessment of the practical feasibility" of SGL imaging given that Earth-like planets have ∼70% cloud cover. The comprehensive benchmark of Turyshev (2026b) reaches a complementary conclusion: a static inverse applied to rotating cloudy data is the wrong estimator; phase-registered robust coadds recover persistent surface structure but at the cost of $\gtrsim$15 registered visits per rotational phase bin, and a "full spin-resolved dynamic retrieval pipeline" is explicitly deferred as a next-stage deliverable. The covariance-aware biosignature framework of Turyshev (2026e) likewise argues that SGL inference must propagate structured error terms rather than treat all residuals as white. In short: the field agrees that the time-domain inverse problem is the gate through which SGL imaging must pass, and no published pipeline yet passes it under Earth-like cloud cover.

This paper attacks that gate directly, with three goals. First (Sections 2–4), we build a transparent, end-to-end simulator—validated against the published photometric and deconvolution benchmarks—in which the scene is dynamic: the planet rotates, the illumination follows the orbit, and an advecting, Ornstein–Uhlenbeck cloud field with adjustable mean cover f_c and decorrelation time τ_c modulates the surface. Second (Section 5), we formulate reconstruction as a regularized generalized least-squares (GLS) inversion of the time-tagged sample stream, in which rotation and illumination are handled exactly by the forward operator and clouds enter statistically through their temporal covariance. We quantify what this estimator family achieves as a function of cloud cover, observing cadence, and model mismatch, against three baselines drawn from the literature: a static Wiener inverse, a phase-binned coadd emulating Turyshev (2026b), and a clouds-as-white-noise GLS emulating the averaging philosophy of Toth (2025). Third (Section 6), we translate the algorithmic findings into mission requirements: observing cadence and swarm size, a ranked table of the five nearest confirmed potentially habitable targets with the celestial placement and dynamics of each focal line, and the propulsion, power, and communications resources needed to operate at $z \gtrsim 550$ AU, drawing on the current architecture literature (Turyshev et al. 2020; Helvajian et al. 2023; Friedman et al. 2024; Turyshev 2026d).

Our headline findings are: (i) rotation and illumination are, by themselves, no longer obstacles—the time-tagged GLS recovers a cloud-free rotating planet essentially perfectly ($r = 0.99$) where prior dynamic reconstructions reached SNR $\lesssim 2$; (ii) clouds dominate the noise budget by an order of magnitude over the solar corona and set the real feasibility limit, in quantitative agreement with the caution of Toth (2025); and (iii) the limit is not fixed: because cloud noise decorrelates in time while the surface signal is rotation-locked, observing cadence is a control knob. At fixed photons and fixed wall-clock time, many short revisits beat few long dwells by factors of ∼4 in recovered correlation. This converts the cloud problem from a physics veto into a requirement on observing architecture — one that directly favors the multi-spacecraft "string of pearls" concept (Helvajian et al. 2023).

2 Validated SGL Photometric Model

2.1 Optical response

We adopt the standard monopole, aperture-averaged description of SGL image formation (Turyshev & Toth 2020; Toth & Turyshev 2021; Turyshev 2026b). A source-plane coordinate ξ maps to the image-plane coordinate $\rho \simeq -(z/z_0)\,\xi$; an Earth-radius planet at $z_0 = 30$ pc observed from $z = 650$ AU projects to

$$D_{\text{img}} = 2R_p\,\frac{z}{z_0} = 1.338\ \text{km}. \tag{1}$$

A telescope of aperture d used as a ring photometer averages the J_0^2 oscillations of the monopole PSF; the effective kernel is

$$K(0) = 1, \qquad K(\rho > 0) \simeq \frac{d}{4\rho}, \tag{2}$$

renormalized so that the mean convolved signal over the projected disk of a reference (albedo 0.3, fully illuminated) planet is unity (Turyshev 2026b). Our discrete kernel follows the $d/4\rho$ tail over the full grid (Fig. 1a). Because K falls only as $1/\rho$, every sample mixes light from the entire disk—the property that couples the temporal variability of any part of the planet into every measurement, and that will dominate the error analysis below.

2.2 Photon rates and noise

We normalize the scene such that a value of 1 corresponds to the disk-mean signal of the reference planet, which maps to the fiducial rates of Turyshev & Toth (2022a): $Q_{\text{exo}} = 8.01\times10^4\,(d/1\,\text{m})^2(650\,\text{AU}/z)^{1/2}(30\,\text{pc}/z_0)(\lambda/1\,\mu\text{m})\ \text{s}^{-1}$ and $Q_{\text{cor}} = 6.20\times10^9\,(d/1\,\text{m})^2(650\,\text{AU}/z)^2(\lambda/1\,\mu\text{m})\ \text{s}^{-1}$ after coronagraphic rejection and annular extraction, with the mean corona assumed subtracted and only its shot noise retained. Each dwell of duration t_s at image-plane position ρ_k yields a Gaussian sample with standard deviation (normalized units)

$$\sigma_k = \frac{\sqrt{(Q_{\text{cor}} + Q_{\text{exo}}\bar{s}_k)\,t_s}}{Q_{\text{exo}} t_s}, \tag{3}$$

where $\bar{s}_k$ is the convolved scene value. For $t_s = 1800$ s this gives $\text{SNR}_\text{C} = 1/\sigma = 43.16$, matching Turyshev (2026b) exactly.

2.3 Deconvolution-penalty validation

As an end-to-end check of the discrete operator we reproduce the noise-amplification ("deconvolution penalty") analysis of

Toth & Turyshev (2021). For a static, fully illuminated planet we convolve, add white noise, deconvolve by Fourier quotient, and measure the ratio $\mathrm{SNR_R}/\mathrm{SNR_C}$ of input noise to reconstructed noise. Figure 2 compares the measurement with the analytic estimate $0.891\,D/(d\sqrt{N})$, where D is the sample spacing and $N = n^2$ the pixel count: for $n = 64$ ($D = 20.9$ m) we measure 0.324 vs. 0.291 predicted; for $n = 128$ ($D = 10.5$ m), 0.106 vs. 0.073. The measured penalties are equal to or slightly milder than the estimate, the same behavior reported by Toth & Turyshev (2021) from their own simulations, and the $n = 32$ case saturates near unity as expected. We conclude the discrete forward operator and noise normalization reproduce the published SGL imaging benchmarks and can be trusted as the substrate for the dynamic experiments that follow.

3 Dynamic Scene and Campaign Model

3.1 Rotating, illuminated planet

The truth surface is a seeded synthetic albedo map (ocean 0.06, continents 0.32 ± 0.10, polar caps up to 0.60; 30% land fraction) generated by thresholded multi-octave Gaussian random fields on a 144×72 longitude–latitude grid (Fig. 1b). Using a procedurally generated planet rather than an Earth image avoids benchmark-specific tuning; the generator and seed are published with the code. The planet rotates with period $P_{\mathrm{rot}} = 24$ h (equatorial viewing geometry, zero obliquity) and is illuminated by its host star with a Lambertian terminator whose sub-stellar point drifts through $\sim 90^\circ$ of orbital phase ($P_{\mathrm{orb}} = 1$ yr) during the campaign. Scenes are rendered to the image-plane disk by orthographic projection with bilinear interpolation, and each dwell averages three sub-exposures so that intra-dwell rotational smear (7.5° per 1800 s dwell) is present in the data but not in the reconstruction operator—a deliberate, realistic model mismatch.

3.2 Stochastic cloud field

Clouds are modeled as an advecting Ornstein–Uhlenbeck (OU) Gaussian random field on the fine grid: spatial correlation length 12°, zonal advection 6° day^{-1}, temporal decorrelation $\tau_c = 4$ d, mapped through a fixed soft threshold to an opacity field with adjustable mean cover $f_c \in [0, 0.72]$. The effective scene is $A_{\mathrm{eff}} = (1-c)A_{\mathrm{surf}} + cA_{\mathrm{cl}}$ with cloud albedo $A_{\mathrm{cl}} = 0.65$. The realized mean covers in our runs are 0.22, 0.38, 0.55(5), and 0.72 for nominal $f_c = 0.25, 0.40, 0.55, 0.70$. This model is deliberately heuristic (as in Toth 2025); its role is to supply statistically stationary, temporally correlated variability whose amplitude and correlation time are known and can be mis-specified in robustness tests (Sec. 5.5).

3.3 Campaign

The fiducial campaign matches the current mission literature (Turyshev & Toth 2022a; Helvajian et al. 2023; Turyshev 2026b): $N_{\mathrm{sc}} = 16$ spacecraft sample an $n \times n = 64 \times 64$ raster ($\Delta_{\mathrm{img}} = 20.9$ m pitch) in boustrophedon order, 16 pixels simultaneously per dwell slot, with dwell $t_s = 1800$ s and $M_p = 16$ complete passes; random inter-pass gaps (0–12 h) emulate calibration and slew overhead and break rotational phase locking. The campaign spans 90.0 d of wall-clock time and accumulates $M_p t_s = 8$ h of photons per pixel ($4096 \times 16 = 65{,}536$ time-tagged samples). Every sample carries its acquisition time; nothing in the pipeline assumes the scene is static.

Cloud-induced signal fluctuations, calibrated from ensembles of rendered scenes, have per-sample standard deviations $\sigma_{\mathrm{cl}} = 0.33$, 0.43, 0.49, and 0.53 (normalized units) for the four cloud covers—i.e., *14–23 times larger* than the coronal noise $\sigma_{\mathrm{cor}} = 0.023$ per 1800 s dwell. This single comparison reframes the SGL noise budget: once the mean corona is subtracted, the photometric war is not against coronal photons but against the planet's own weather.

4 Reconstruction Methods

4.1 Time-domain inversion (TDI)

Let $\mathbf{s} \in \mathbb{R}^{n_s}$ be the (static) surface albedo map on a 72×36 reconstruction grid (coarser than the truth grid, so the inversion never sees the generating basis). Each sample k, taken at time t_k and image-plane pixel p_k, has the linear forward model

$$y_k = \mathbf{f}_k^{\mathsf{T}}\mathbf{s} + \varepsilon_k, \qquad \mathbf{f}_k^{\mathsf{T}} = \mathbf{k}_{p_k}^{\mathsf{T}}\,\Pi(t_k), \tag{4}$$

where $\Pi(t_k)$ renders the map to the disk at the rotation phase and illumination of time t_k (known ephemeris) and $\mathbf{k}_{p_k}$ is the SGL kernel row at the sampled pixel. The error ε_k contains coronal shot noise (variance σ_k^2, Eq. 3) and the cloud perturbation. Because the mean cloud field is uniform in expectation, its climatological bias is removable by the affine correction $\hat{\mathbf{s}} = (\hat{\mathbf{s}}_{\mathrm{raw}} - f_c A_{\mathrm{cl}})/(1-f_c)$; the fluctuating part has per-sample variance σ_{cl}^2 and temporal covariance well approximated by $\exp(-|\Delta t|/\tau_c)$ for revisits of the same pixel.

The TDI estimator is regularized generalized least squares,

$$\hat{\mathbf{s}} = \arg\min_{\mathbf{s}} \|\mathsf{C}^{-1/2}(\mathbf{y} - \mathsf{F}\mathbf{s})\|^2 + \lambda_{\mathrm{eff}}\|\mathsf{L}\mathbf{s}\|^2, \tag{5}$$

with L the spherical-grid Laplacian and λ_{eff} set by a fixed dimensionless weight $\lambda = 3 \times 10^{-3}$ scaled by $\mathrm{tr}(\mathsf{F}^{\mathsf{T}}\mathsf{C}^{-1}\mathsf{F})/\mathrm{tr}(\mathsf{L}^{\mathsf{T}}\mathsf{L})$ (chosen once on a calibration seed and frozen). We consider two covariance options and one projection:

- *Temporal whitening:* C is block diagonal over image-plane pixels; for the M_p visits of one pixel, $\mathsf{C}_p = \sigma_{\mathrm{cl}}^2 \exp(-|t_i - t_j|/\tau_c) + \mathrm{diag}(\sigma_k^2)$.
- *White:* $\mathsf{C} = \mathrm{diag}(\sigma_k^2 + \sigma_{\mathrm{cl}}^2)$—the assumption that clouds average like white noise, i.e., the estimator-side formalization of Toth (2025).
- *Slot deflation:* because the $1/\rho$ kernel wings mix the whole disk into every sample, the cloud state at each dwell

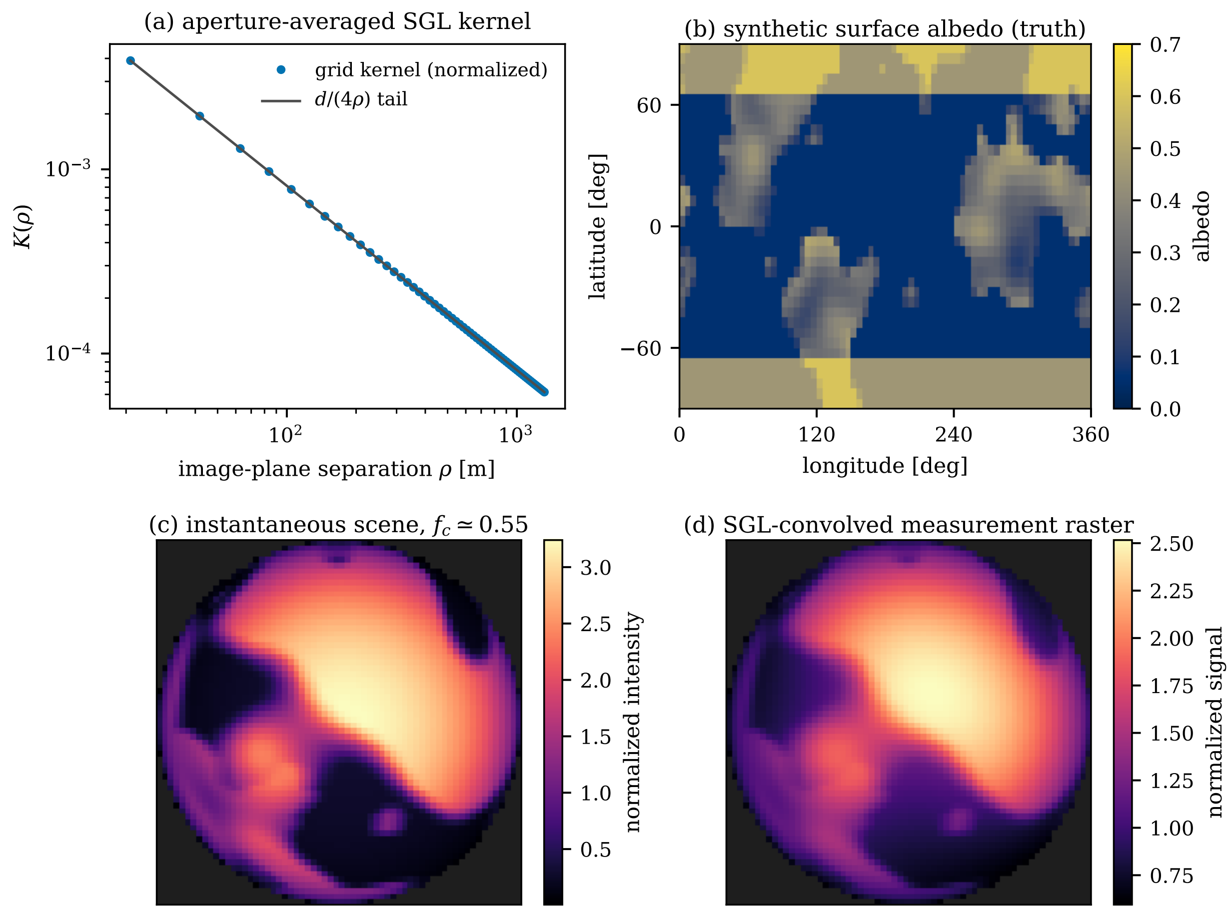

Figure 1: Model overview. (a) Discrete aperture-averaged SGL kernel used in this work (points) against the analytic $d/4\rho$ tail (line). (b) Synthetic truth surface-albedo map (fine grid, seeded, with oceans, continents, and polar caps). (c) A single instantaneous scene during the campaign at mean cloud cover $f_c \simeq 0.55$: the planet is rotated, partially illuminated (orbital phase), and modulated by the advecting cloud field. (d) The same scene after convolution with the SGL kernel: the measurement raster a spacecraft would sample. Note the severe blur and the loss of small-scale contrast.

slot injects an error component common to the $N_{\rm sc}$ simultaneous samples. We optionally project out the per-slot mean from data and operator (sacrificing $1/N_{\rm sc}$ of the information), which removes this globally correlated mode exactly.

The normal equations ($n_s = 2592$ unknowns, 6.6×10^4 samples) are accumulated in blocks and solved directly; a full reconstruction takes $\sim$10 s on a laptop-class CPU. Below, "TDI" denotes the deflated, temporally whitened variant unless stated otherwise; the ablation between covariance options is part of the results.

4.2 Baselines

B1, static Wiener: average all visits per pixel and apply a Wiener inverse of the kernel—the classical static pipeline (Toth & Turyshev 2021), known to be the wrong estimator for dynamic scenes (Turyshev 2026b). *B2, phase-binned coadd:* bin samples into 8 rotational phase bins, Wiener-deconvolve each binned raster, and back-project the deconvolved views onto the surface grid with illumination weighting—a transparent emulation of the phase-registered robust-coadd strategy of Turyshev (2026b). *B3, clouds-as-white-noise GLS:* Eq. (5) with the white covariance and no deflation. All map-space methods receive the same climatological de-biasing, and all tunable constants were fixed on a calibration seed excluded from the results.

4.3 Metrics

Reconstructions are compared with the truth map (block-averaged to the reconstruction grid) over latitudes $|b| \le 60°$ using the structural similarity index (SSIM), Pearson correlation r, and normalized RMS error. Three seeds per configuration control the realization scatter of clouds and noise.

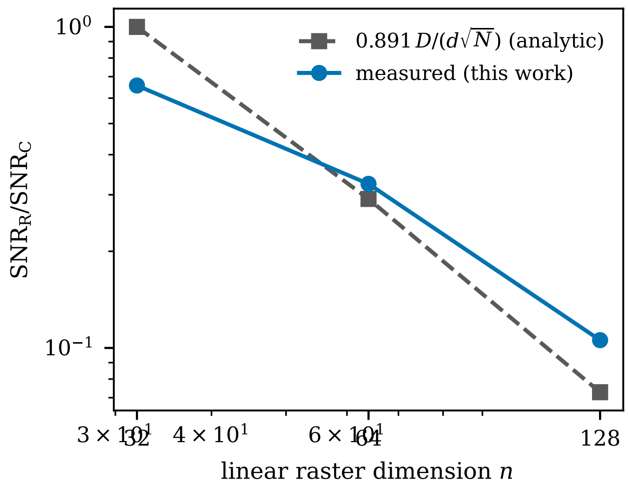

Figure 2: Validation of the discrete operator: measured deconvolution noise penalty vs. the analytic estimate of Toth & Turyshev (2021). Measured penalties are equal to or slightly milder than predicted, matching the behavior reported in that work; the n=32 point saturates toward unity.

Table 1: Map fidelity vs. mean cloud cover (fiducial campaign; mean over three seeds).

Method	f_c=0	0.22	0.38	0.56	0.72
		Pearson r			
TDI (this work)	0.990	0.572	0.432	0.334	0.263
B3 white-noise GLS	0.991	0.543	0.402	0.308	0.239
B2 phase-binned coadd	0.669	0.328	0.215	0.129	0.052
		SSIM			
TDI (this work)	0.918	0.281	0.172	0.109	0.065
B3 white-noise GLS	0.930	0.272	0.165	0.102	0.060
B2 phase-binned coadd	0.311	0.132	0.081	0.045	0.011

Realized mean covers quoted; nominal $f_c = 0$, 0.25, 0.40, 0.55, 0.70. All methods receive identical data, the same climatological de-biasing, and constants frozen on a held-out calibration seed.

5 Results

5.1 Rotation and illumination are solved problems

For a rotating, orbit-illuminated but cloud-free planet, TDI recovers the surface with $r = 0.990 \pm 0.001$ and SSIM $= 0.92$ (Fig. 3, top center): coastlines, continental shapes, and polar caps are essentially exact at the resolution of the reconstruction grid. The phase-binned coadd baseline reaches only $r = 0.67$ (SSIM $= 0.31$) on identical data, and the static Wiener image is unusable, consistent with the C3 branch of Turyshev (2026b). This is the first headline result: once every sample is time-tagged and the rotation/illumination ephemeris is folded into the forward operator, the "temporal aliasing" of the raster scan costs almost nothing in the noise-dominated regime—the SNR $\lesssim 2$ ceiling of earlier dynamic reconstructions (Toth & Turyshev 2023) was a property of the estimator, not of the information content of the data.

5.2 Clouds set the feasibility limit

Table 1 and Figure 4 quantify the collapse of image fidelity with mean cloud cover at fixed observing resources. Two features stand out. First, the degradation is steep: r falls from 0.99 (cloud-free) to 0.57 at 22% cover and 0.33 at 56% cover; SSIM falls faster still, because small-scale structure is erased first. Second, the ranking of methods is uniform: TDI $>$ white-noise GLS $>$ phase-binned coadd $\gg$ static, with TDI improving r over the white-noise assumption by $\sim$5–10% (relative) across the sweep. The joint-inversion family as a whole outperforms the coadd strategy by factors of 2–3 in r at every cloud cover.

These numbers place the qualitative pessimism of Toth (2025) on a quantitative footing while revising its implication. At the observing budget both studies assume (single-visit-scale statistics; here 8 h of photons per pixel over 16 visits), Earth-like cloud cover indeed destroys small-scale surface recovery: no estimator we tested reaches SSIM > 0.15 at $f_c \gtrsim 0.55$. But the gross bimodal geography (land–ocean dichotomy, polar caps) remains detectable at $r \simeq 0.3$, and—critically—the loss is not photon-starvation but *correlated nuisance variance*, which observing strategy can attack (Sec. 5.4).

5.3 Structure of the cloud-noise and covariance ablation

The measured residuals $\mathbf{y} - \mathsf{F}\mathbf{s}_{\text{true}}$ match the calibrated cloud model precisely (standard deviation 0.489 vs. 0.486 predicted at $f_c = 0.55$), confirming that the fidelity collapse is driven by the modeled cloud statistics rather than by operator error. Three structural facts explain the modest margin between the covariance options. (i) The $1/\rho$ kernel injects a globally correlated error mode: the instantaneous disk-integrated cloudiness perturbs *every* sample of that dwell slot coherently. Slot deflation removes this mode and is included in TDI; it accounts for most of TDI's advantage at the fiducial cadence. (ii) The per-pixel temporal covariance is weak at the fiducial revisit interval (5.6 d $\gtrsim \tau_c = 4$ d; adjacent-visit correlation $\simeq$0.25), so temporal whitening has little to exploit—and at fast cadence (Sec. 5.4), where visits *are* correlated, whitening trades variance suppression against effective sample loss nearly evenly: the white variant is modestly better at $M_p \geq 32$ (e.g., $r = 0.58$ vs. 0.54 at $M_p = 64$), while deflation+whitening wins at slow cadence. (iii) The residual cloud error is spatially smooth and enters through the same operator as the surface, so no diagonal-in-any-basis covariance can fully separate the two; a full spatio-temporal cloud covariance (or explicit cloud-state estimation) is the natural next step and we quantify its headroom as the gap between the $f_c = 0.55$ and cloud-free rows of Table 1.

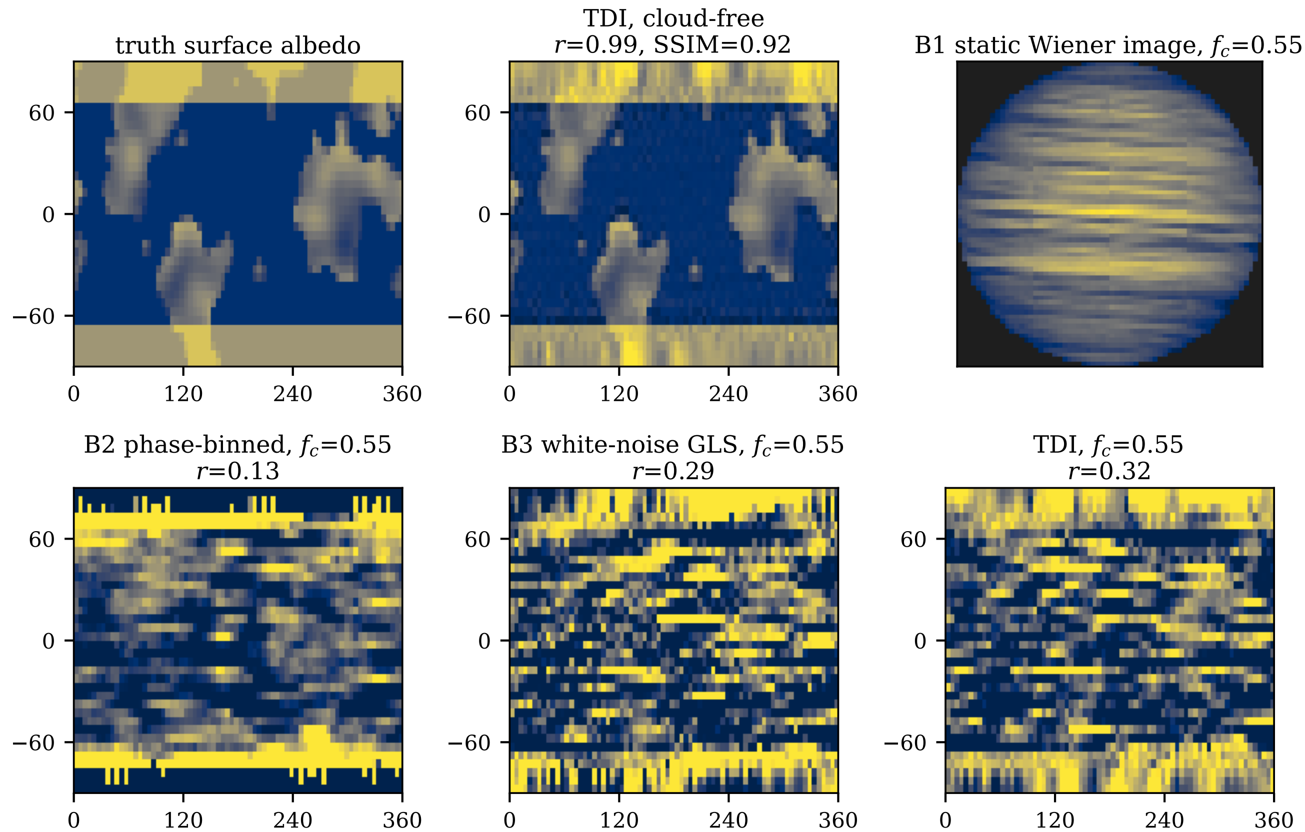

Figure 3: Reconstruction gallery for the fiducial 90-day, 16-spacecraft campaign. Top: truth map; TDI reconstruction of the rotating cloud-free planet ($r = 0.99$); static Wiener image (B1) of the $f_c = 0.55$ data, illustrating why a static inverse fails. Bottom, all at $f_c \simeq 0.55$: phase-binned coadd (B2), clouds-as-white-noise GLS (B3), and TDI. Even at 55% cloud cover TDI recovers the gross continental dichotomy, but small-scale fidelity is lost (Sec. 5.2).

5.4 The observing-cadence law

Because the surface signal is rotation-locked and deterministic while cloud noise decorrelates on τ_c, the *distribution* of a fixed photon budget in time is a free design parameter with large leverage. We repeated the $f_c = 0.55$ experiment at fixed total photons per pixel ($M_p t_s = 8$ h) and near-fixed wall-clock time (86–101 d), trading dwell time against number of passes: $(t_s, M_p) = (7200\,\mathrm{s}, 4) \rightarrow (450\,\mathrm{s}, 64)$. Figure 5 shows the result: recovered correlation rises monotonically from $r = 0.15$ (4 long dwells) to $r = 0.58$ (64 short revisits; 0.54 for the deflated variant, 0.58 for the white variant, cf. Sec. 5.3)—a factor of $\sim$4 at *identical* photon cost. Coronal noise per visit grows as $t_s^{-1/2}$ but remains negligible against σ_{cl} even at $t_s = 450$ s ($\mathrm{SNR_C} = 21.6$ per visit); meanwhile each extra pass samples a fresher cloud state and shortens intra-dwell rotational smear. The practical rule for cloud-noise-limited SGL imaging is therefore: *revisit as fast as coronal noise allows, down to* $\sigma_{\mathrm{cor}} \sim \sigma_{\mathrm{cl}}/\sqrt{M_p}$. Extrapolating the measured trend, reaching $r \gtrsim 0.8$ at Earth-like cloud cover requires several hundred registered visits per pixel—i.e., either year-class campaigns with tens of spacecraft or 90-day campaigns with $\mathscr{O}(10^2)$ spacecraft, a direct quantitative sizing input for string-of-pearls architectures (Helvajian et al. 2023). This cadence dividend is invisible to single-visit noise budgets and is, to our knowledge, quantified here for the first time.

5.5 Robustness and the rotation-period requirement

TDI requires a cloud climatology $(\sigma_{\mathrm{cl}}, \tau_c)$ and the rotation ephemeris a priori. Figure 6 shows single-seed sensitivity scans at $f_c = 0.55$: misestimating σ_{cl} by a factor of 2 in either direction changes SSIM by $<1\%$, and misestimating τ_c by a factor of 4 changes it by $<30\%$ (with a mild preference for shorter assumed τ_c). The estimator is thus insensitive to the details of the assumed cloud statistics—only their order of magnitude matters. The rotation period is different: a fractional error of 10^{-4} is harmless, but 10^{-3} (phase drift $\sim 30^\circ$ over the campaign) destroys the reconstruction (SSIM $0.106 \rightarrow 0.024$). SGL imaging of a cloudy planet therefore levies a hard precursor requirement: *the target's rotation period must be known to* $\sim 10^{-4}$ ($\sim$10 s for a 24 h rotator), attainable from long-baseline photometric monitoring of rotational modulation prior to and during

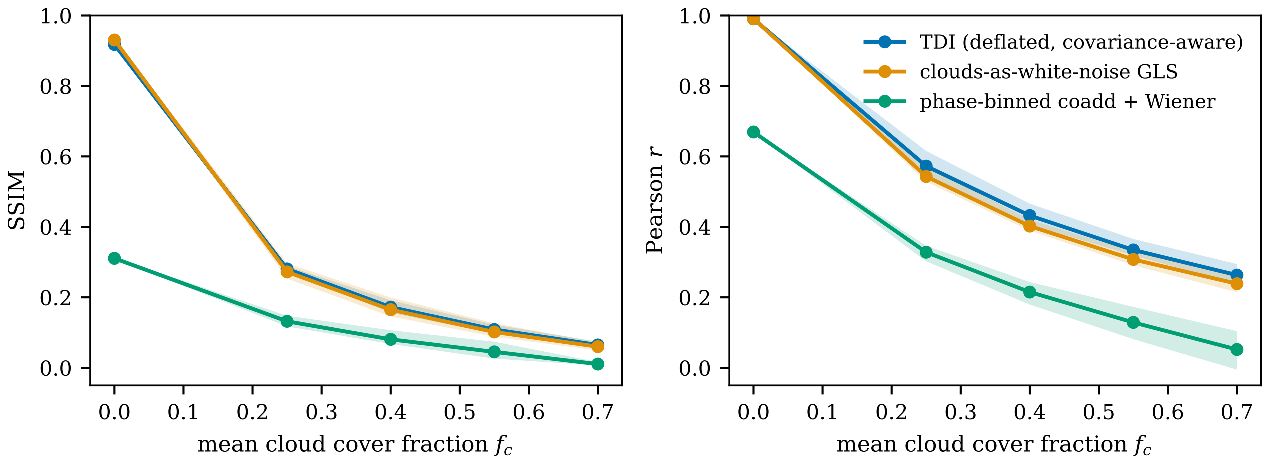

Figure 4: Reconstruction fidelity vs. mean cloud cover for the three map-space methods (mean and seed-to-seed range over three seeds). Clouds—not coronal photon noise—control SGL image quality at fixed observing resources.

the mission, and refinable in-flight by maximizing map self-consistency over a period grid.

6 From Algorithm to Mission

The lens is target-specific: the focal line extends anti-parallel to the target direction, and a spacecraft cruising outward at $\sim$150 km s^{-1} has no meaningful capability to retarget (a 1° change of line-of-sight at 650 AU corresponds to an 11.3 AU lateral displacement; Turyshev 2026c,d). Target selection, trajectory, and onboard resources are therefore inseparable from the imaging algorithm requirements derived above. This section assembles them into a coherent mission picture.

6.1 The five best targets and their focal-line placements

Table 2 evaluates the five nearest *confirmed* planets that current radial-velocity work places in or near their stars' habitable zones, plus one solar-type alternative, as SGL imaging targets. For each planet we list the adopted physical parameters (discovery references in the table notes), and compute: the projected image-cylinder diameter $D_{\rm img}$ at $z = 650$ AU; the fiducial photon rate $Q_{\rm exo}$ scaled from the 30 pc benchmark ($Q \propto S R_p z_0^{-1}$ at fixed z, d, λ; Turyshev & Toth 2022a) and the resulting per-dwell SNR$_{\rm C}$(1800 s); the celestial placement of the focal line (the antipode of the target's J2000 position, with its ecliptic latitude); and the image-plane dynamics—the radius $r_{\rm img} = a_p(z/z_0)$ and speed $v_{\rm img} = v_{\rm orb}(z/z_0)$ of the circular track the planetary image sweeps around the host star's focal line each planetary year, together with the Δv a spacecraft would expend to track the image continuously for 90 days. Radii for these non-transiting planets are estimated from $M \sin i$ via a rocky mass–radius relation $R \propto M^{0.28}$ (Chen & Kipping 2017); all photometric quantities therefore carry the (unknown) $\sin i$ and albedo systematics, and should be read as ranking metrics rather than forecasts.

Several mission-shaping conclusions follow.

1. Proxima Centauri b is the photometric and geometric optimum: 15 times the fiducial photon rate (SNR$_{\rm C} = 659$ per 1800 s dwell) and a 31.5 km image cylinder whose generous pixel spacing essentially eliminates the deconvolution penalty (at $n = 128$, $0.891D/d\sqrt{N} > 1$; even a 1024×1024 megapixel raster incurs only a factor $\sim$0.03 penalty, so megapixel imaging of Proxima b is photon-cheap: SNR$_{\rm R} \simeq 18$ per 1800 s-per-pixel pass with 16 spacecraft in $\sim$3.7 yr of cumulative dwell). Its liabilities are dynamical: the 11.2 d orbit sweeps the planetary image around the stellar focal line at 114 m s^{-1} (track radius 1.76×10^4 km), so continuous tracking would cost 5.8 km s^{-1} per 90 d—infeasible for smallsats. Proxima b therefore forces the picket strategy analyzed by Turyshev & Toth (2022b): spacecraft hold station near the stellar focal line and sample the image during periodic sweeps, which our cadence result actually favors (many short, phase-diverse encounters). The focal line at 02$^{\rm h}$30$^{\rm m}$, +62°41′ (ecliptic latitude +45°) requires a high-inclination departure.

2. Ross 128 b offers the best photon-dynamics compromise among confirmed planets: SNR$_{\rm C} = 576$, half Proxima's tracking burden, a quiet M4 host (Bonfils et al. 2018), and a focal line lying *in* the ecliptic ($\beta = +0.5°$), the cheapest departure geometry.

3. GJ 1061 d is the most temperate confirmed target ($S = 0.69 S_\oplus$) with moderate dynamics, but its focal line at +61° ecliptic latitude is the most expensive to reach.

4. Teegarden c orbits the least active host (M7V; Dreizler et al. 2024), minimizing stellar-contamination systematics in the Einstein ring, at the cost of the lowest photon rate (SNR$_{\rm C} = 129$; still ample—corona-limited dwell budgets scale as SNR$_{\rm C}{}^{-2}$ and remain subdominant to cloud noise). Its focal line also lies in the ecliptic.

5. GJ 273 b (Luyten b) is the most massive and thus most

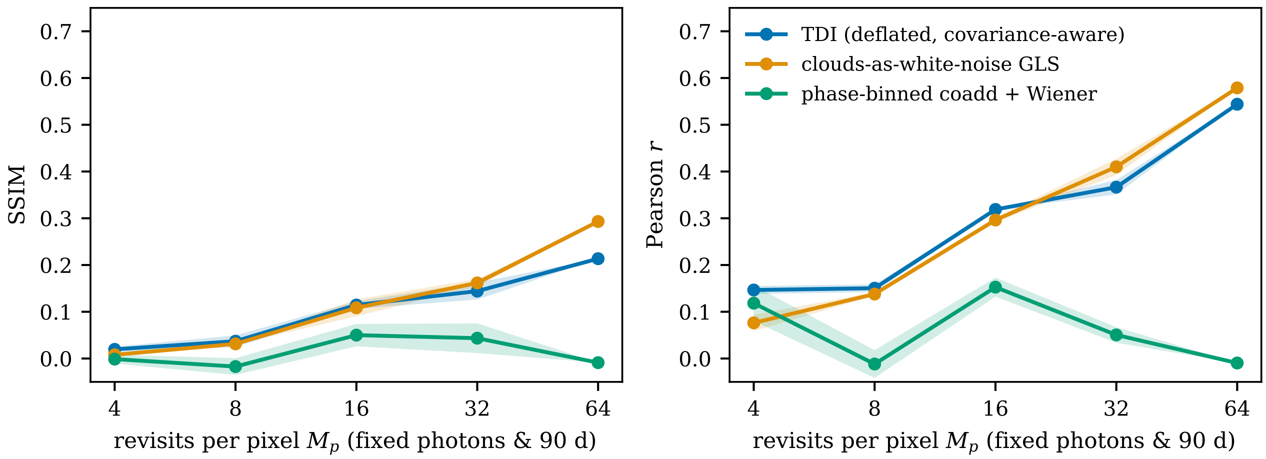

Figure 5: The observing-cadence law at $f_c \simeq 0.55$: fidelity vs. revisits per pixel M_p at fixed photons per pixel ($M_p t_s = 8$ h) and near-fixed wall-clock time. Many short revisits beat few long dwells by factors of $\sim$4 in r, because revisits sample independent cloud states while the surface signal remains rotation-locked. The coadd baseline (B2) cannot exploit the extra visits.

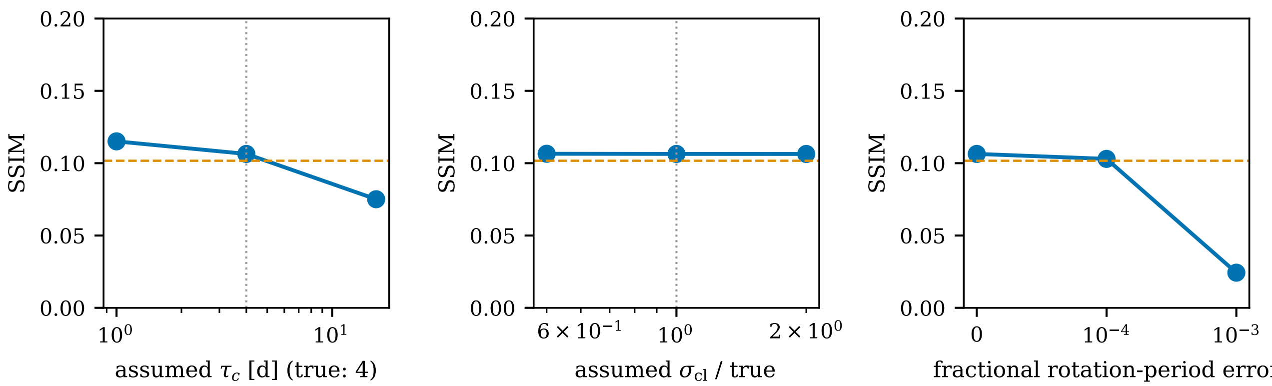

Figure 6: Robustness of TDI at $f_c \simeq 0.55$ (single seed). Left, center: SSIM under mis-specified cloud decorrelation time and amplitude (dotted lines: truth; dashed orange: white-noise GLS reference). Right: sensitivity to rotation-period error—10^{-4} is tolerable, 10^{-3} is fatal, setting a precursor-knowledge requirement.

likely to retain a substantial atmosphere; it combines high $\mathrm{SNR_C} = 486$ with the lowest tracking cost among the confirmed five (1.3 km s^{-1} per 90 d, thanks to its wider 18.65 d orbit).

The solar-type alternative. If τ Ceti e is confirmed, it becomes arguably the best imaging target of all: the highest photon rate in the table ($\mathrm{SNR_C} = 901$), and—because its habitable zone lies at 0.54 AU rather than 0.05 AU—image-plane dynamics *fifty times gentler* than Proxima b (0.11 km s^{-1} per 90 d), compatible with continuous tracking by modest electric propulsion. This illustrates a general and previously underappreciated selection rule: *M-dwarf habitable zones are photon-rich but dynamically expensive; solar-type habitable zones are dynamically cheap.* Rotation periods (Sec. 5.5) are the second selection axis: tidally synchronized M-dwarf planets ($P_{\rm rot} = P_{\rm orb}$, known to 10^{-5} from RV ephemerides) automatically satisfy the 10^{-4} period requirement, converting a liability of M-dwarf targets into an asset—though synchronization also fixes the illuminated hemisphere, halving the mappable surface.

Since the mission is irrevocably single-target (Turyshev 2026c), the final selection must await precursor characterization (atmosphere detection, rotation state, cloud statistics from disk-integrated variability) by ELT-class and Habitable Worlds Observatory-era facilities; the ranking above identifies where that precursor effort buys the most mission risk reduction.

6.2 Getting there: deep-perihelion sailcraft

Reaching $z = 650$ AU within a 20 yr program implies a mean radial speed of 32.5 AU yr^{-1} (154 km s^{-1})—far beyond chemical or gravity-assist propulsion, whose flight-heritage maximum is 3–4 AU yr^{-1} (Turyshev 2026d). The schedule-credible path is photonic: drop toward the Sun, deploy a large low-mass

Table 2: The five nearest confirmed potentially habitable planets (and one solar-type alternative) as SGL imaging targets. Observing geometry evaluated at $z = 650$ AU, $d = 1$ m, $\lambda = 1\,\mu$m.

Planet	Host (SpT)	z_0 (pc)	$M \sin i$ ($M_\oplus$)	R_p[a] ($R_\oplus$)	S ($S_\oplus$)	P (d)	$D_{\rm img}$ (km)	SNR$_{\rm C}$ (1800 s)	Focal direction[b] (J2000)	$\beta_{\rm ecl}$ (deg)	Δv_{90}[c] (km s^{-1})
Proxima Cen b	Proxima (M5.5V)	1.30	1.07	1.02	0.65	11.19	31.5	659	$02^{\rm h}30^{\rm m}$, $+62°41'$	+44.8	5.8
Ross 128 b	Ross 128 (M4V)	3.38	1.35	1.09	1.38	9.87	12.9	576	$23^{\rm h}48^{\rm m}$, $-00°48'$	+0.5	2.9
GJ 1061 d	GJ 1061 (M5.5V)	3.67	1.64	1.15	0.69	13.03	12.6	280	$15^{\rm h}36^{\rm m}$, $+44°31'$	+60.8	1.7
Teegarden c	Teegarden's Star (M7V)	3.83	1.11	1.03	0.37	11.41	10.8	129	$14^{\rm h}53^{\rm m}$, $-16°53'$	−0.3	1.7
GJ 273 b	Luyten's Star (M3.5V)	3.80	2.89	1.35	1.06	18.65	14.2	486	$19^{\rm h}27^{\rm m}$, $-05°14'$	+16.5	1.3
τ Cet e[d]	τ Ceti (G8.5V)	3.60	3.93	1.47	1.71	162.9	16.3	901	$13^{\rm h}44^{\rm m}$, $+15°56'$	+24.8	0.11

Image-plane sweep speeds $v_{\rm img}$ are 114, 51, 39, 35, 44, and 31 m s^{-1} from top to bottom. [a]Estimated from $M \sin i$ via $R \propto M^{0.28}$ (Chen & Kipping 2017); none of these planets transit. [b]Antipode of the host's J2000 position (arcminute precision); the usable focal segment extends along this direction from $z \simeq 548$ AU outward, with 650–900 AU the practical operating range (Turyshev 2026d). $\beta_{\rm ecl}$ is the ecliptic latitude of the focal direction. [c]Propellant cost of continuously tracking the planetary image's orbital sweep for 90 days, $\Delta v_{90} = a_{\rm img} \times 90$ d; picket-style sampling (Turyshev & Toth 2022b) reduces this dramatically. [d]Radial-velocity candidate, not confirmed; included as the nearest potentially habitable planet of a solar-type star. Parameters: Anglada-Escudé et al. (2016); Faria et al. (2022) (Proxima b); Bonfils et al. (2018) (Ross 128 b); Dreizler et al. (2020) (GJ 1061 d); Zechmeister et al. (2019); Dreizler et al. (2024) (Teegarden c); Astudillo-Defru et al. (2017) (GJ 273 b); Feng et al. (2017) (τ Cet e). Instellations from the discovery papers.

sail near perihelion r_p, and exit hyperbolically with

$$v_\infty \simeq \sqrt{\frac{2\mu_\odot \beta}{r_p}}, \qquad \beta = \frac{p_0}{\sigma_{\rm tot}\, g_\odot(1\,{\rm AU})}, \tag{6}$$

where $\mu_\odot = GM_\odot$, $p_0 = 9.08\ \mu{\rm N\,m^{-2}}$, and $\sigma_{\rm tot}$ is the *total* system areal density including payload and—critically—the non-solar power source (Turyshev 2026d). The quantitative requirements are stringent: at $r_p = 0.05$ AU ($10.7\ R_\odot$), $v_\infty \simeq 105$ km s^{-1} (30 yr class) requires $\sigma_{\rm tot} \simeq 4.9$ g m^{-2}, and $v_\infty \simeq 155$ km s^{-1} (20 yr class) requires $\sigma_{\rm tot} \simeq 2.3$ g m^{-2}; a 100 kg sailcraft at $\sigma_{\rm tot} = 4.9$ g m^{-2} implies a 2×10^4 m^2 ($\sim$140 m square) sail. Thin-membrane equilibrium at $r_p = 0.05$ AU reaches $\sim$740 K (for absorptivity 0.05, emissivity 0.8), and a representative 50 kg, 10^4 m^2 point design achieving 21 AU yr^{-1} from $r_p = 0.025$ AU saw 950 K (Helvajian et al. 2023). Sail deployment at 10 m scale is flight-proven, but deep-perihelion material systems are at low technology readiness; the 25–40 yr-class regime ($r_p \sim 0.05$–0.1 AU, $\sigma_{\rm tot}$ of a few g m^{-2}) is the realistic near-term envelope, with sub-20 yr access gated by a focused materials-and-deployment qualification program (Turyshev 2026d).

The alternatives trade schedule for capability. Nuclear electric propulsion with a 20 t stage, 800 kg payload, $I_{\rm sp} = 9000$ s and specific mass $\alpha_{\rm tot} = 10$–20 kg kW$_e^{-1}$ reaches 650 AU in 27–33 yr while delivering hundreds of kW$_e$ for operations; sub-20 yr NEP requires either $\alpha_{\rm tot} \lesssim 3$ kg kW$_e^{-1}$ or a hybrid architecture in which an Oberth-maneuver injection stage (e.g., nuclear-thermal, firing at $r_p \sim 0.02$ AU) supplies $v_0 \simeq 50$–70 km s^{-1} before the NEP cruise (Turyshev 2026d). For the swarm-heavy architectures our cadence law favors (Sec. 5.4), the sail-first roadmap is the natural match: each $\lesssim$100 kg "pearl" rides its own sail, launches are staggered at 1–2 yr intervals so that later pearls carry improved technology (Helvajian et al. 2023), and the large sail area is retained at the focal region for reuse as a solar occulter (Turyshev 2026d).

6.3 Power and communications at 500–1000 AU

Beyond $\sim$500 AU solar flux is 4×10^5–8×10^5 times weaker than at 1 AU: photovoltaics are useless and every watt must be nuclear (Turyshev 2026d). The conventional option, the Multi-Mission Radioisotope Thermoelectric Generator, supplies $\sim$110 W$_e$ at $\lesssim$45 kg—a 20–50% mass fraction for a 100–200 kg sailcraft that directly inflates $\sigma_{\rm tot}$ and, through $v_\infty \propto \sigma_{\rm tot}^{-1/2}$, the trip time; indeed a 1 kW$_e$ system at 100 kg kW$_e^{-1}$ adds sail area equivalent to an 80 m-radius disk just to carry its own power supply (Turyshev 2026d). The architecture developed for exactly this problem is the Atomic Planar Power for Lightweight Exploration (APPLE) concept: modular $10 \times 10 \times 1.7$ cm tiles integrating ^{238}Pu heat sources with thermoelectric converters and radiation-hard solid-state batteries, each tile delivering $\geq$1.2 W$_e$ continuously plus 5 Wh of storage for $\geq$15 yr, with decay heat keeping the battery warm without electrical heaters (NASA/Aerospace Corp. 2022; Turyshev et al. 2020). Tiled power scales from watts to hundreds of watts at areal mass compatible with sailcraft, and the battery buffers the peak loads of our observing concept—short high-duty-cycle dwells, image-plane maneuvering, and burst downlink.

The data volume demanded by time-domain imaging is modest. Each sample is a time-tagged photometric integral ($\lesssim$32 bytes); the fiducial campaign produces 6.6×10^4 samples ($\sim$2 MB), and even a megapixel, 64-visit Proxima campaign totals $\sim$2 GB of raw photometry. Optical-communication architectures studied for the string-of-pearls concept—inter-pearl laser relays hopping down the line of spacecraft, with the innermost "caboose" pearl closing the link to Earth—project tens to hundreds of kb s^{-1} from 500–800 AU (Helvajian et al. 2023), i.e., days per full raster: downlink is not a driver. The binding resource conclusions are instead (i) per-pearl power of order 10^2 W$_e$ (tiled RPS), (ii) transverse image-plane mobility budgets of 0.1–5.8 km s^{-1} per 90 d depending on target (Table 2), and (iii) the swarm multiplicity itself, which our cadence law sets at $\mathcal{O}(10^2)$ registered visits per pixel for Earth-like cloud

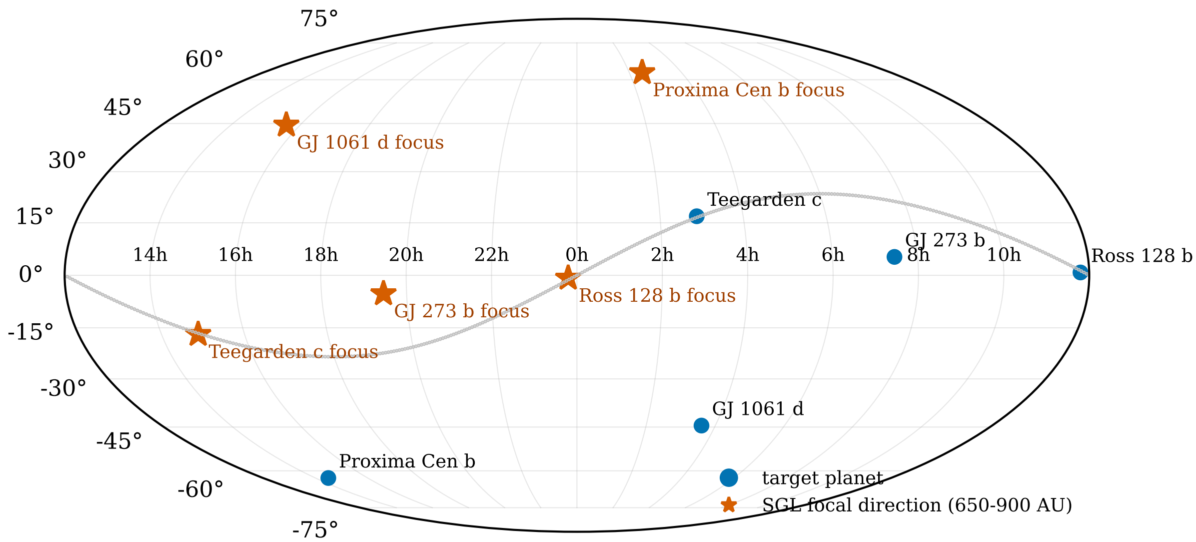

Figure 7: Sky placement (equatorial Mollweide) of the five ranked targets (blue circles) and their SGL focal directions (orange stars)—the antipodal points toward which the spacecraft must cruise. The gray curve is the ecliptic. Note that the Ross 128 and Teegarden focal lines lie almost exactly in the ecliptic plane, while GJ 1061's focus is at $+61^\circ$ ecliptic latitude, with consequences for launch geometry, navigation-field stars, and zodiacal foregrounds.

cover—achievable as 16 craft $\times$ years or 10^2 craft $\times$ 90 d.

7 Discussion and Limitations

Our simulations are deliberately transparent rather than exhaustive, and several simplifications bound the claims. (i) We use the monopole, aperture-averaged SGL response; the solar quadrupole breaks the PSF into caustics that increase the deconvolution burden (Turyshev & Toth 2021), and extending TDI with a J_2-dependent kernel library is the most important optical upgrade. (ii) The corona enters as Gaussian shot noise with its mean assumed subtracted; structured coronal residuals and calibration drifts—shown by Turyshev (2026b) to demand sub-ppm effective flat-fielding—are not modeled. (iii) The cloud field is a single-layer stochastic opacity screen without radiative transfer, phase-angle dependence, or climate structure; it is calibrated, however, so that the estimator never receives information unavailable to a real mission, and the robustness scans show factor-of-few climatology errors are benign. (iv) The geometry is a single equatorial, zero-obliquity case; oblique and pole-on geometries redistribute (but do not eliminate) the rotational information TDI exploits. (v) Navigation is perfect: pointing jitter and image-plane ephemeris error enter as an additional convolution and their coupling to the 10^{-4} rotation-period requirement deserves a dedicated study. (vi) Truth maps live on a finer grid than the reconstruction, dwell-smear is present in data but not operator, and tuning constants were frozen on held-out seeds—standard inverse-crime mitigations, though only flight-like simulation chains (Turyshev 2026b) can retire the concern fully. (vii) Our per-pixel temporal covariance is the simplest member of the covariance-aware family; the measured gap to the cloud-free ceiling (factor $\sim$3 in r at Earth-like cover) quantifies the prize for full spatio-temporal cloud modeling, e.g., joint estimation of a low-order cloud state per epoch or GCM-informed priors, in the spirit of Turyshev (2026e).

These caveats do not touch the two structural results—that rotation-aware joint inversion removes the temporal-aliasing penalty, and that cloud noise, being temporally correlated nuisance variance rather than photon noise, is purchasable with revisits rather than photons. Both follow from the linearity of the measurement and the separation of timescales ($P_{\rm rot} \ll \tau_c \ll T_{\rm camp}$), not from details of our cloud model.

8 Conclusions

We built a validated, open, end-to-end simulator of exoplanet imaging with the solar gravitational lens—reproducing the published kernel, photon-rate, and deconvolution-penalty benchmarks—and used it to attack the time-domain inverse problem that the current literature identifies as the concept's leading unsolved challenge. Our conclusions:

1. Time-domain inversion (regularized GLS on time-tagged samples with the rotation/illumination ephemeris in the forward operator) recovers a rotating, cloud-free planet with $r = 0.99$, SSIM $= 0.92$—removing the SNR $\lesssim 2$ barrier of earlier dynamic reconstructions and demonstrating that planetary rotation, per se, costs SGL imaging almost nothing.

2. Clouds dominate the noise budget, exceeding coronal shot noise per sample by factors of 14–23 for covers of 22–72%.

At single-campaign budgets, fidelity collapses to $r \simeq 0.26$–0.33 at Earth-like cover for every estimator tested, quantitatively confirming published caution—but the gross land–ocean dichotomy survives.

3. Cloud noise is correlated, and that is exploitable. A globally correlated mode (injected by the $1/\rho$ kernel wings) is removable by per-slot deflation; the temporal correlation defines an observing-cadence law: at fixed photons and wall-clock, 64 short revisits outperform 4 long dwells by a factor $\sim$4 in r. Cadence, not collecting area, is the control knob for cloudy targets, and it directly sizes multi-spacecraft swarms: $\mathscr{O}(10^2)$ registered visits per pixel for high-fidelity mapping under Earth-like clouds.

4. The method needs the rotation period to $\sim 10^{-4}$ and only order-of-magnitude cloud climatology—a clean, achievable precursor requirement set.

5. Among the five nearest confirmed potentially habitable planets, Proxima b is photon-optimal but dynamically expensive (image sweeps at 114 m s^{-1}; 5.8 km s^{-1} per 90 d of continuous tracking), Ross 128 b offers the best compromise with an in-ecliptic focal line, and GJ 273 b minimizes tracking cost; a confirmed τ Ceti e would dominate the ranking outright. Focal-line placements, photon rates, and tracking budgets for all targets are tabulated for mission design.

6. The transportation and resource stack is consistent with the current architecture literature: deep-perihelion sailcraft ($\sigma_{\rm tot} \simeq 2.3$–$4.9$ g m^{-2} at $r_p = 0.05$ AU for 155–105 km s^{-1}), tiled radioisotope power ($\geq$1.2 W$_e$ per $10 \times 10 \times 1.7$ cm APPLE tile), and optical relay downlink at tens of kb s^{-1}—with the swarm multiplicity demanded by our cadence law emerging as the single strongest architectural driver.

The time-domain problem, in short, is not a veto but a design driver: it converts “can the SGL image a living, breathing planet?” into specific, testable requirements on cadence, swarm size, precursor ephemerides, and covariance modeling. All code, seeds, and data needed to reproduce every number in this paper accompany the manuscript.

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A Solver implementation

The normal equations of Eq. (5) are assembled per image-plane pixel: for pixel p with visit set $\mathscr{V}_p$ ($|\mathscr{V}_p| = M_p$), the local covariance $\mathsf{C}_p = \sigma_{\rm cl}^2 \mathsf{T}_p + \mathrm{diag}(\sigma_k^2)$, $(\mathsf{T}_p)_{ij} = e^{-|t_i - t_j|/\tau_c}$, is Cholesky-factored ($M_p \times M_p$) and applied to the corresponding rows of F and entries of $\mathbf{y}$; whitened blocks are accumulated into $\mathsf{A} = \mathsf{F}^{\mathsf{T}}\mathsf{C}^{-1}\mathsf{F}$ and $\mathbf{b} = \mathsf{F}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{y}$ in single precision and solved in double precision. Slot deflation subtracts from each sample (and operator row) the mean over the $N_{\rm sc}$ simultaneous samples of its dwell slot, and rescales the white-noise term by $\sqrt{1 - 1/N_{\rm sc}}$; it is an orthogonal projection, so the deflated GLS remains unbiased for $\mathbf{s}$ up to the removed global mode, which is restored by the climatological de-bias. The regularization weight is dimensionless: $\lambda_{\rm eff} = \lambda\, \mathrm{tr}(\mathsf{A})/\mathrm{tr}(\mathsf{L}^{\mathsf{T}}\mathsf{L})$ with $\lambda = 3 \times 10^{-3}$ for all results. Building the $65{,}536 \times 2592$ operator takes $\sim$3 s and each solve $\sim$10 s (two CPU cores).

B Reproducibility

The archive accompanying this paper contains `code/sglsim.py` (physics, scene, campaign, estimators; $\sim$600 lines), `code/run_experiments.py` (checkpointed experiment driver), `code/targets.py` (Table 2 and Fig. 7), and `code/make_figures.py`. All random processes are seeded; the seeds used here are 7 (truth map), 2026 (campaign), 11–13 (noise/cloud realizations), 99/199 (held-out calibration), and 999 (climatology probes). Every number quoted in the text and tables regenerates from `run_experiments.py` in $\sim$40 core-minutes.

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